

Architecture Selection Justification

Chose Layered Architecture for the Student Startup Crowdfunding Platform System.

Architectural Choice

The Crowdfunding Platform is organized into three layers: a Presentation Layer (User Interface Component), a Business Layer (Campaign Manager, Payment Processing, and Notification Service Components), and a Data Layer (Campaign Database Component). The Campaign Manager sits at the center of the Business Layer and mediates all communication between the presentation, payment, notification, and data components, consistent with the standard Layered Architecture style.

Two Reasons

- 1. Clear separation between student-facing features and financial logic:** Students will frequently browse campaigns, create new ones, and track funding progress, and this presentation-layer activity needs to change often (new pages, UI experiments) without touching how pledges are processed or stored. A Layered Architecture keeps the interface free to evolve independently of the payment and data logic beneath it.
- 2. A natural, ordered pledge workflow:** A pledge must be validated, charged through Payment Processing, recorded in the Campaign Database, and then followed by a Notification Service update to the backer and campaign owner. A Layered Architecture enforces this as a controlled top-down sequence through the Campaign Manager, rather than allowing components to call each other in unpredictable orders, which matters for a small student-built system without a distributed infrastructure team to manage that complexity.

Security Advantage

Because the User Interface Component can only reach the Business Layer through the Campaign Interface, it has no direct path to Payment Processing or the Campaign Database. Every pledge request must pass through the Campaign Manager, which acts as a single, controlled checkpoint that validates and authorizes requests before any payment or student financial data is touched — reducing the surface exposed to a malicious or malformed request coming from the client.

Performance Benefit

Since all components run within a single deployed system rather than as independently networked services, calls between layers (e.g., Campaign Manager to Payment Processing or the Campaign Database) avoid the network latency a distributed style would add. This keeps pledge confirmation and campaign-page updates fast for students browsing and backing projects, without the operational cost of running and scaling separate services for a platform at student-project scale.